

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_003258 | Base: main-v1.0

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Decision: NO_GO | Confidence: 53%

CHANGE REQUEST

vorrei cambiare il peso da 33.8kg a 36kg

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Mass exceeds accommodation envelope

Subsystem: structure | Req: L1-MAS-001 | Type: constraint_violation

The proposed mass increase from 33.8 kg to 36 kg violates the platform's accommodation envelope of 35 kg, leading to potential structural and thermal issues.

[MAJOR] ISSUE-002 — Potential impact on mechanical subsystems

Subsystem: mechanical | Req: None | Type: reanalysis_required

Increasing the payload mass may affect the structural integrity and vibration characteristics of the mechanical subsystem, potentially requiring reanalysis or additional testing.

[MAJOR] ISSUE-003 — Thermal management constraints

Subsystem: thermal | Req: None | Type: reanalysis_required

The increased mass may affect thermal dissipation and heat distribution, potentially impacting the thermal subsystem's ability to maintain temperature stability.

[MINOR] ISSUE-004 — Power consumption considerations

Subsystem: power | Req: None | Type: reanalysis_required

The additional mass may require more power for operation, potentially impacting the power subsystem's ability to meet the required power budget.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The proposed mass increase may necessitate re-evaluation of shock response and additional testing to ensure compliance with the launcher's structural requirements.

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM

Reason: The increased mass may affect thermal management, requiring a re-evaluation of the thermal model and additional testing.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	41%	WARN	Thermal management constraints	■ MODIFIED
mechanical	68%	WARN	Potential impact on mechanical subsystems	■ MODIFIED
structure	56%	FAIL	Mass exceeds accommodation envelope	■ MODIFIED
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-MAS-001 — verification evidence invalidated, re-analysis required